Web Dev Basics 1

CS571: Building User Interfaces

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Before Class

Clone today's example to your machine!

Learning Objectives

1. Obtain a broad understanding of web programming.
2. Understand the essentials of JavaScript.
3. Be able to use JavaScript in web programming.
4. Know of other popular tools.

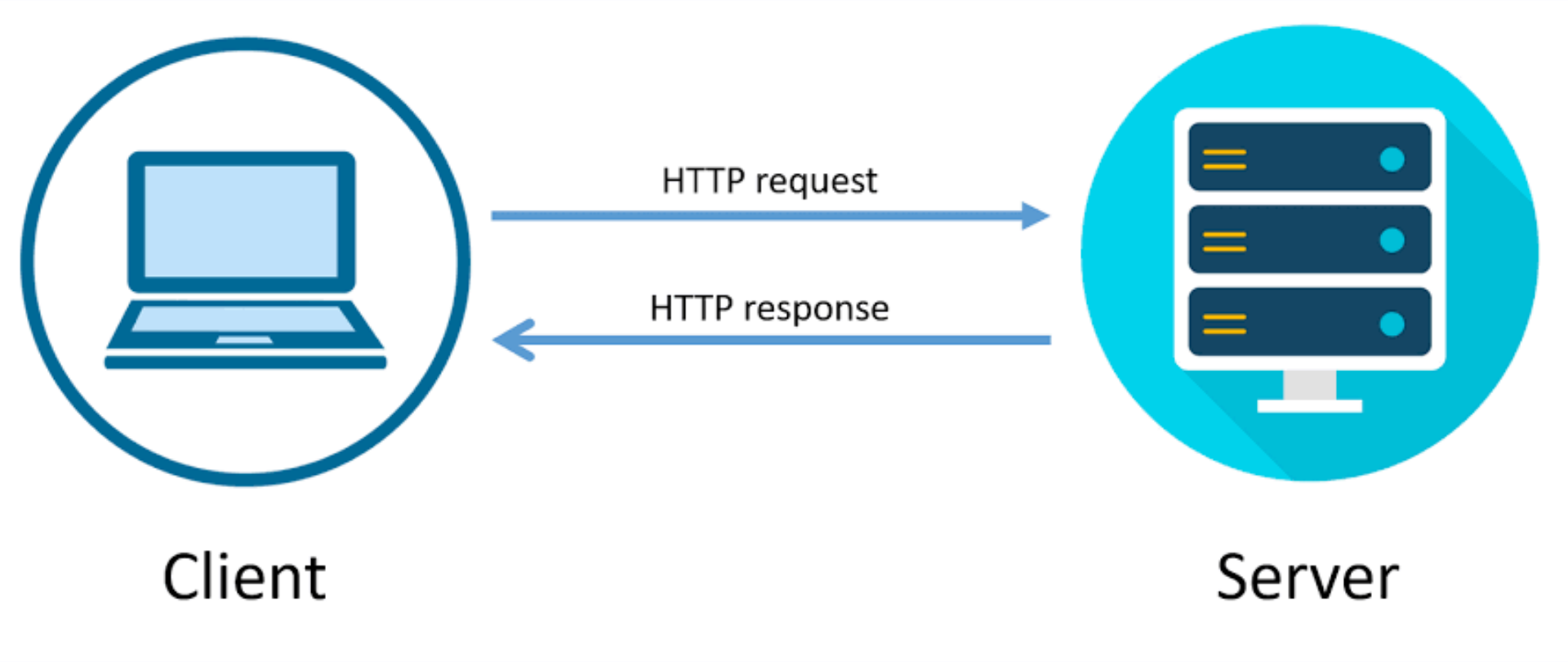
Web Programming

This is not a comprehensive introduction to web programming, so these are great additional resources:

- MDN Web Docs
- W3 Schools

Periodically, we will also use StackBlitz or CodePen for code snippets.

Web Communication



Medium

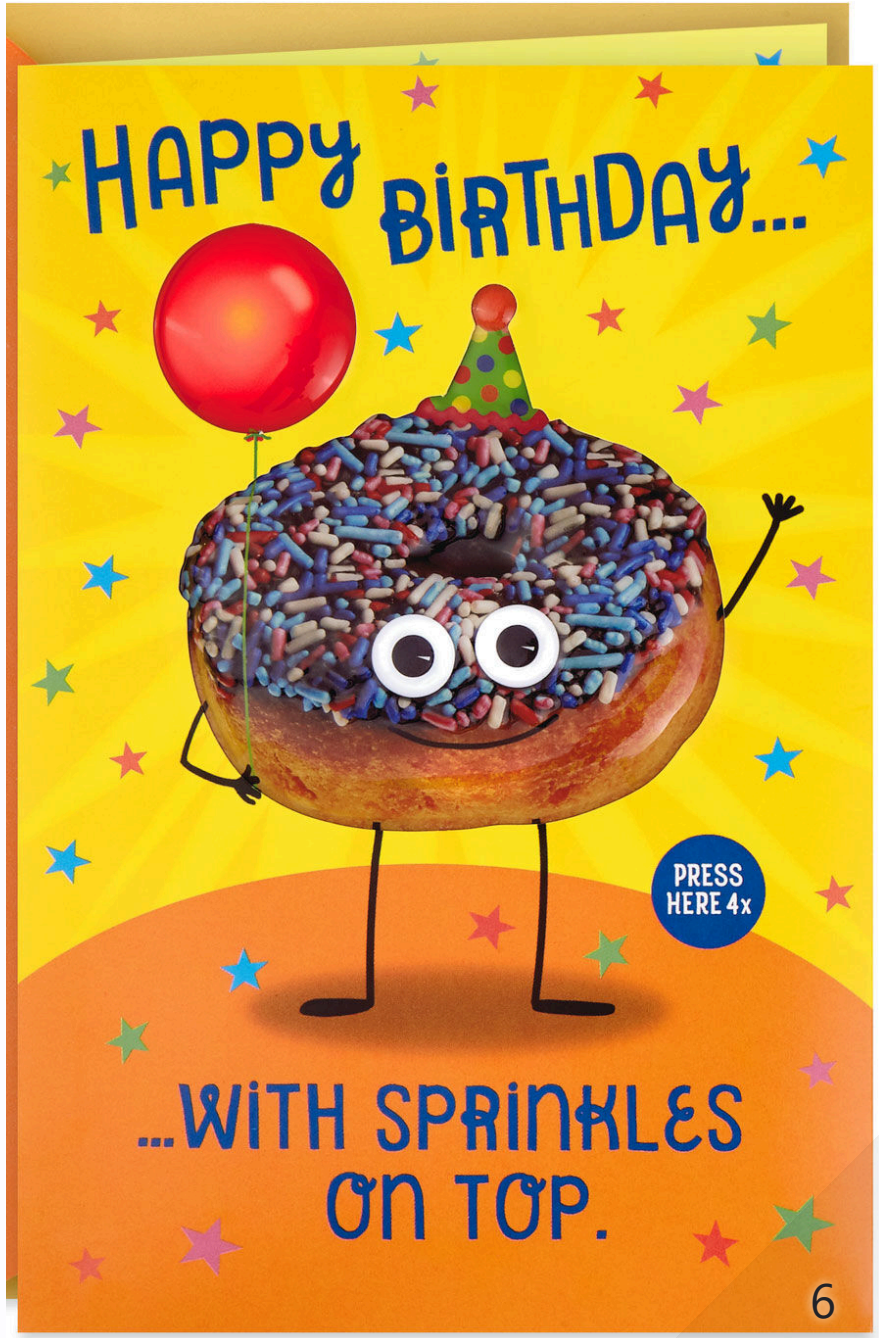
A Website

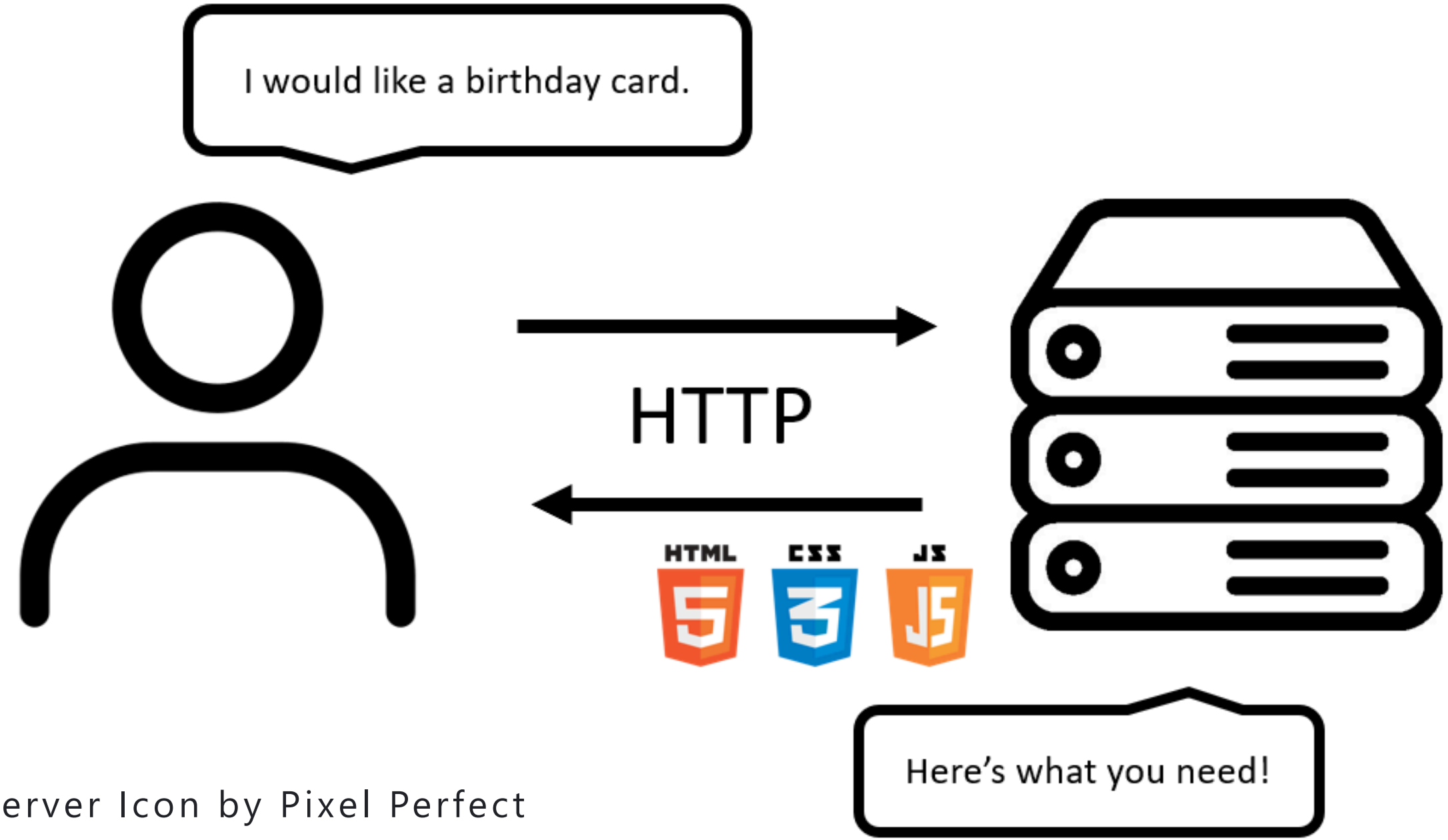
It's like a birthday card!

Concept	Behavior
HTML	Structure
CSS	Design
JS	Behavior

How does it get delivered? HTTP!

Personal Website Codepen





Server Icon by Pixel Perfect

HTML

Hyper-Text Markup Language

Defines the structure of the webpage.

Made up of a series of "elements" (sometimes referred to as "nodes") defined by "tags".

```
<h1>Hello world!</h1>
<div>
  <p>My favorite foods are...</p>
</div>
```

HTML

Each element can have 0-to-many attributes defined as key-value pairs in its tag.

Each element can have a unique `id` attribute - useful for CSS and JS!

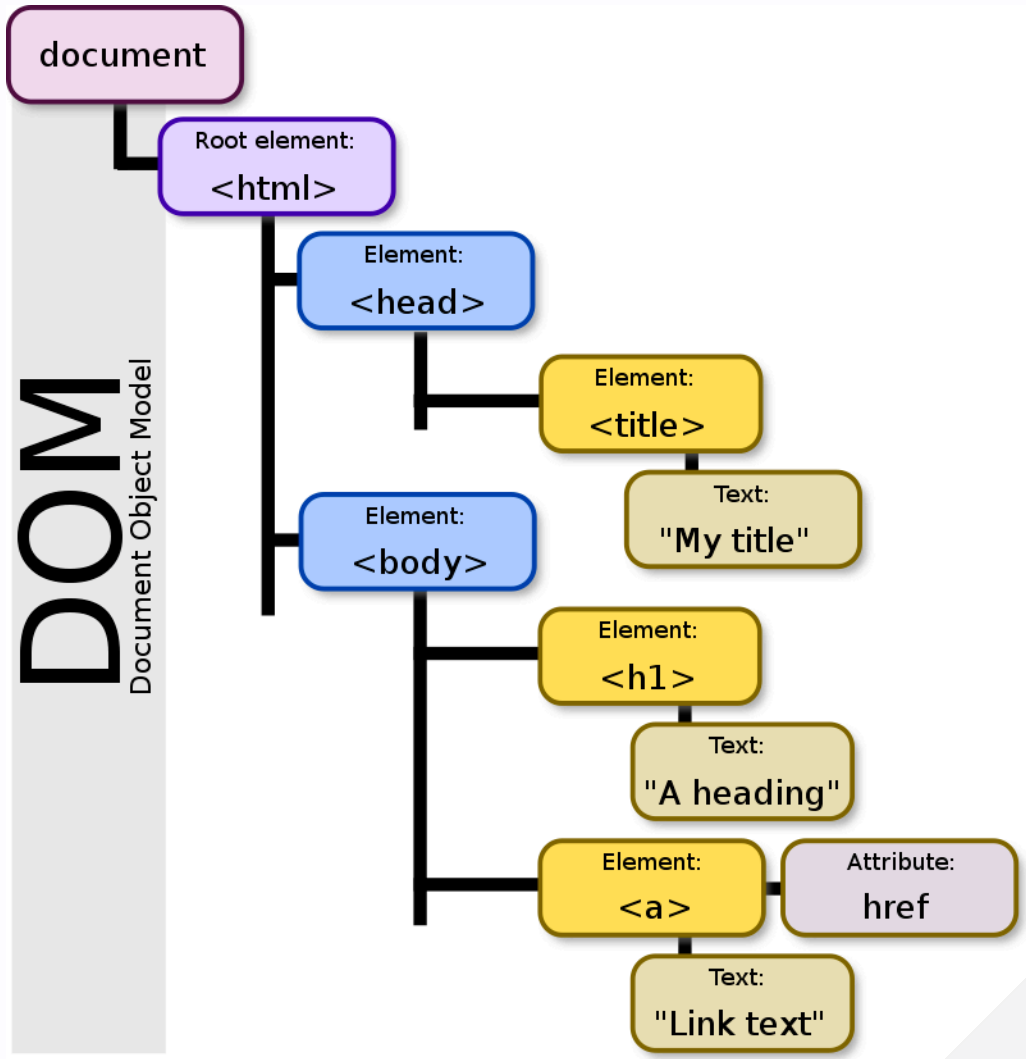
```
<h1 id="intro-txt" style="font-size: 99">Hello world!</h1>
<p>Visit my <a href="http://example.com/">other website</a></p>
<img id='cole' src="http://example.com/me.png" alt="A picture of Cole"/>
```

HTML

Notice the structure?

We call this the Document Object Model (DOM).

This will be very useful later!



HTML

Tags we should know...

- `html` : encapsulate entire html document
- `head` : include metadata, such as CSS and JS scripts
- `body` : main HTML content

```html
<html lang="en">
    <head></head>
    <body>
      <p>My HTML would go here...</p>
    </body>
</html>
```

HTML

Tags we should know...

- `h1` through `h6` : headers
- `p` paragraph text
- `strong` and `em` (instead of `b` and `i`): connotation
- `a` : anchor text (links)
- `img` : images (always include an `alt` attribute!)

HTML

Tags we should know…

- `br` : linebreak
- `ul` , `ol` , `li` : lists
- `table` , `thead` , `tbody` , `tr` , `th` , `td` : tables
- `div` : divisions (sections)

HTML

Tags we should know...

- `form` : encapsulates many inputs
- `label` : label an input (always include a `for` attribute)
- `button` : buttons
- `input` : user input (text, radio, button)

Badger Recipes

Just HTML

CSS

Cascading Style Sheets

Defines the styles to be used.

Uses a cascading hierarchy of selectors and decoration boxes consisting of key-value pairs.

```
p {
  color: blue;
  font-size: 16px; /* prefer em or rem; e.g. 1rem */
}
```

CSS

`` ` `` applies to all HTML elements with a particular tag.

`#` applies to all HTML elements with a particular id.

`.` applies to all HTML elements with a particular class.

You can mix and match these!

You can decorate many elements with comma-seperated lists!

CSS

Example	Description
#firstname	Selects the element with id="firstname"
.intro	Selects all elements with class="intro"
p.intro	Selects `<p>` elements with class="intro"
*	Selects all elements
p	Selects all `<p>` elements
div, p	Selects all `<div>` and `<p>` elements

CSS

There are three ways to include your own css...

1. Inline
2. Internal
3. External

When doing CSS, prefer external over inline!

CSS Inline

```
<div style="background-color: powderblue">
  <h1 style="font-size: 4rem">Badger</h1>
  <ul>
    <li style="background-color: orange">a heavily built omnivorous nocturnal mammal</li>
    <li style="color: red">of the weasel family</li>
    <li>typically having a gray and black coat.</li>
  </ul>
  <p style="text-align: right">From the <strong>Oxford</strong> dictionary</p>
</div>
```

StackBlitz

CSS Internal

```
<style>
  #main-div {
    background-color: green;
    margin: 1rem;
  }
  li {
    color: yellow;
    font-size: 1.25rem;
  }
  .right-text {
    text-align: right;
  }
</style>
```

StackBlitz

CSS External

Same as internal but with styling in an external file, e.g. `styles.css`. Include stylesheet in HTML with...

```
<link rel="stylesheet" href="styles.css"/>
```

StackBlitz

CSS Cascading

A complicated process, but generally, the style closest to the element wins! You can override this by adding `!important`

Use your dev tools.

Read the notes.

```
element.style {
}

.uw-pe.uw-pe-latest_posts .uw-posts-listing .uw-post .u
a, .uw-pe.uw-pe-latest_posts .uw-posts-listing .uw-post
  margin-top: -1rem;
  vertical-align: top;
}

a, ins {
  text-decoration: ▶ none;
}

a {
  background-color: transparent;
⚠ -webkit-text-decoration-skip: ~~objects~~;
  color: #0479a8;
}

a, b, em, i, small, strong {
  line-height: inherit;
}

a, abbr, acronym, address, applet, big, blockquote, bod
caption, cite, code, dd, del, dfn, div, dl, dt, em, fie
h6, html, iframe, ins, kbd, label, legend, li, object,
strike, strong, sub, sup, table, tbody, td, tfoot, th,
☑ border: ▶ 0;
☑ font-family: inherit;
```

Badger Recipes

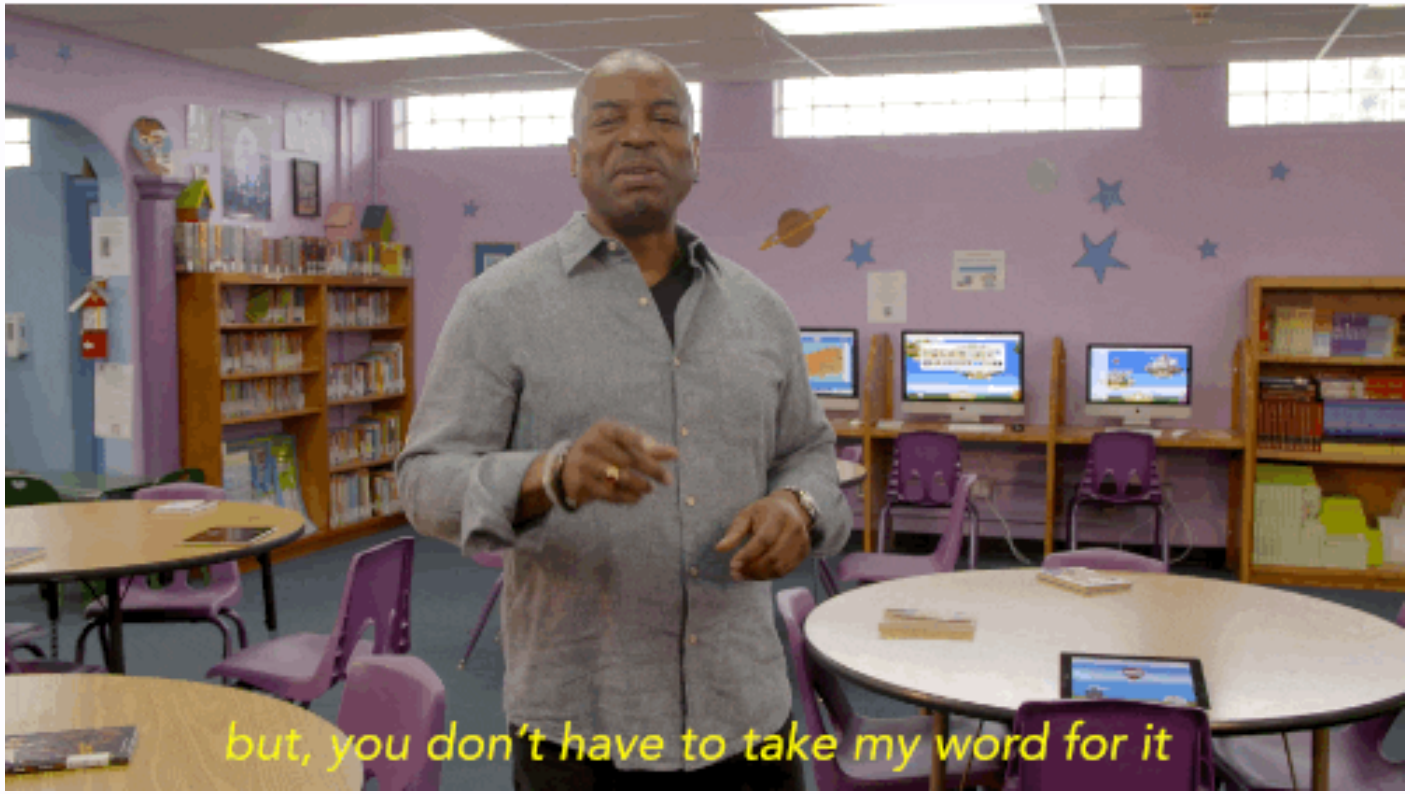
Just HTML and CSS

JavaScript History

- Developed by Netscape Communications (Brendan Eich) in 1995.
  - It was designed in 10 days.
- A "glue language" for HTML.
- Mocha > LiveScript > JavaScript.
- Specifications are ECMAScript (e.g. "ES").

Try It!

Browser > F12 > Console



Variables

JS is a dynamic, loosely-typed language. The data type is inferred from the declaration and can be changed.

Variables are *containers* that hold data.

There are 7 standard data types: numbers, string, boolean, null, undefined, symbol, object. The first 6 are considered *primitive* and are stored on the *stack*. Object is considered *complex* and stored on the *heap*.

Variable Declarations

Variables can be declared with `var`, `let`, or `const`.

Keyword	Scope	Re-assignable?	Preferred?
var	function	yes	no
let	block	yes	yes
const	block	no	yes

`let` and `const` were introduced in ES6. They are the preferred ways of declaring a variable.

Variable Examples

Consider the following block of code...

```
let age = 27;
const name = "Ashley";
var hasCar = false;
```

Can we perform `age = 28`? Yes! We use `let`.

Can we perform `name = "Carl"`? No! We use `const`.

Is `hasCar` declared correctly? Yes, but we should use `let` instead of `var`.

Variable Examples

Is this block of code correct?

```
let name = "Ahmed";
console.log("My name is " + name);
name = 27;
console.log("My age is " + name);
name += 1;
console.log("My age a year later is " + name);
```

It's not *good* code, but it is *correct* code! Variables can change types during runtime.

Determining Data Types

We can query the data type at runtime using `typeof`.

```
let foo = "Charles";
console.log(typeof foo);
foo = 1932;
console.log(typeof foo);
foo = true;
console.log(typeof foo);
```

```
string
number
boolean
```

Conditionals

Conditionals allow the code to make decisions and carry out different actions.

Three types:

1. `if`, `else if`, and `else` statements
2. `switch` statements
3. ternary operators `evalExpr ? trueExpr : falseExpr`

Conditionals

Any value that is not `false`, `undefined`, `null`, `0`, `NaN`, or `""` returns `true`. Why is this useful?

```
var currentMember = "Alice";
let textContent = "?";
if (currentMember) {
  textContent = "View Profile";
} else {
  textContent = "Sign Up";
}
console.log(textContent)
```

```
'View Profile'
```

Comparison and Logical Operators

Operator	Meaning
`===` and `!==`	strong comparison
`==` and `!=`	weak comparison
`<` and `>`	less/greater than
`<=` and `=>`	less/greater than or equal to
`&&`	and
`||`	or

Objects

Objects are collections of data defined using key-value pairs. These are sometimes referred to as "attributes" or "properties".

```
let teachingAssistant = {
   firstName: "Alice",
   lastName: "Smith",
   age: 24
};
console.log(teachingAssistant);
```

```
{firstName: 'Alice', lastName: 'Smith', age: 24}
```

JSON to JS Object

Wait... Haven't we seen that before?... Close!

```
{
 "firstName": "Cole",
 "lastName": "Nelson",
 "roles": ["student", "faculty"]
}
```

JSON

⟷

```
const instructor = {
 firstName: "Cole",
 lastName: "Nelson",
 roles: ["student", "faculty"]
}
```

JS Object

Object Properties

There are two different notations to access object properties.

```
console.log(teachingAssistant.lastName);
console.log(teachingAssistant["firstName"]);
```

```
'Smith'
'Alice'
```

Arrays

An array is a variable that contains multiple elements.

```
let fruits = ["apple", "banana", "coconut"];
fruits[0] = "apricot";
console.log(fruits);
fruits.push(17);
console.log(fruits);
```

```
['apricot', 'banana', 'coconut']
['apricot', 'banana', 'coconut', 17]
```

Notice! They don't have to be the same type.

Looping

For Loops

```
for (let i = 0; i < 10; i++) {}  // typical for loop
for (let attr in course) {}      // loop through object properties
for (let item of arr) {}         // loop through array contents
```

While Loops

```
while(cond) {}                   // typical while loop
do {} while(cond);               // guarantees atleast one execution
```

You can use `break` (exit) and `continue` (skip).

Functions

A procedure that includes a set of statements that performs a task or calculates a value.

Function Definition (Hoisted)

```
function fToC(temp) {
  return (temp - 32) * 5/9;
}
fToC(77)
```

```
25
```

Other Ways to Declare Functions

Function Expression (NOT Hoisted)

```
const fToC = function(temp) {
  return (temp - 32) * 5/9;
}
```

Arrow Function (NOT Hoisted)

```
const fToC = (temp) => {
  return (temp - 32) * 5/9;
}
```

Your turn!

Write a JavaScript function that takes an array of numbers and computes their sum, e.g.

```
[1, 3.2, -4.0, 5] => 5.2
```

... can you do it as a function, function expression, and arrow function?

Tip: Write your code in notepad and paste it into your browser's developer console to test!

JavaScript and the Web

How does this all go together?

JavaScript and the Web

Just like CSS, we can include JavaScript...

1. Inline JS
2. Internal JS
3. External JS

Inline JavaScript

The JavaScript is included inside of the HTML element.

index.html

```
<html>
  <h1>Welcome to my webpage!</h1>
  <button onclick="console.log('hello world!')">Click Me!</button>
</html>
```

Internal JavaScript

The JavaScript is included inside of the HTML.

index.html

```
<html>
  <h1>Welcome to my webpage!</h1>
  <script>
    console.log("hello world!");
  </script>
</html>
```

External JavaScript

The JavaScript is included from outside of the HTML.

index.html

```html
<html>
  <h1>Welcome to my webpage!</h1>
  <script src='app.js'></script>
</html>
```

app.js

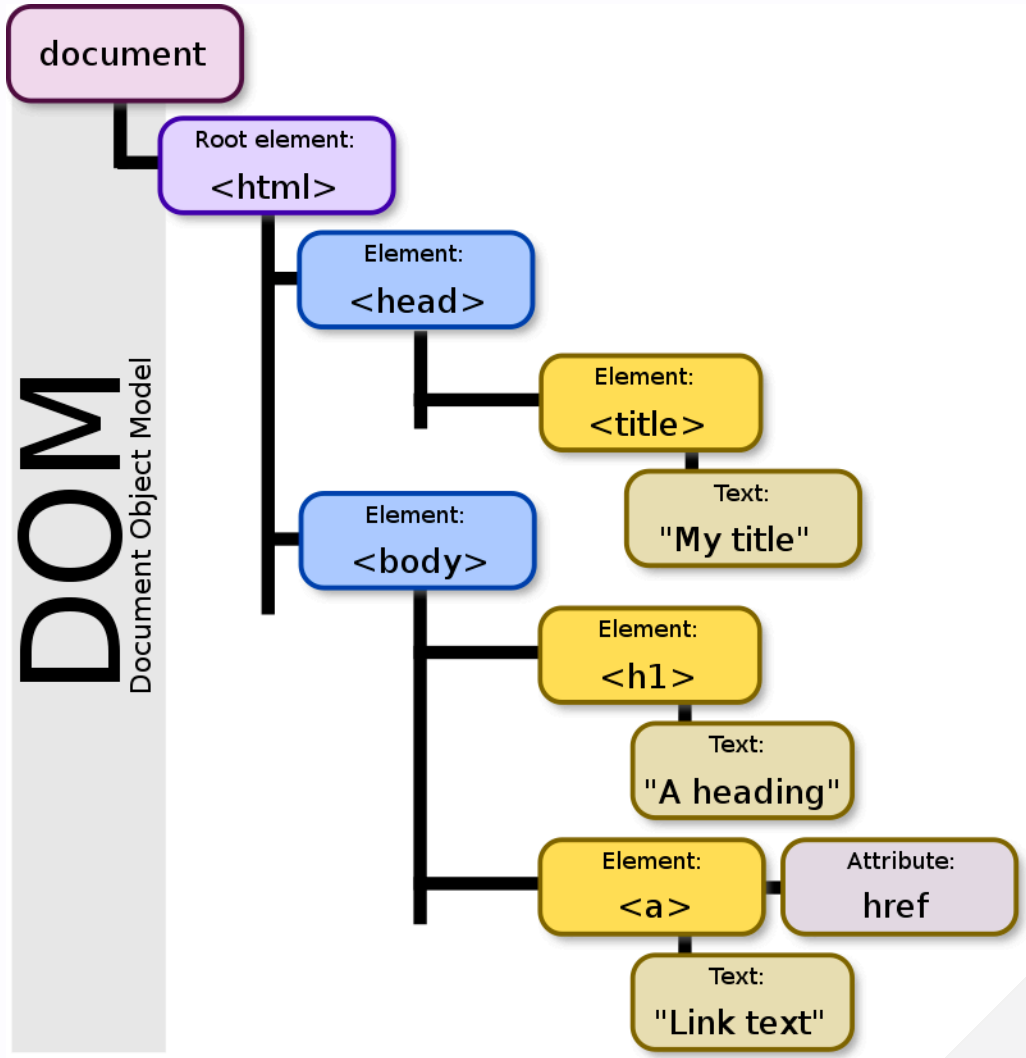
```javascript
console.log("hello world!");
```

Document Object Model (DOM)

HTML is just a tree, where each element is a node!

We use JavaScript to manipulate this tree.

Document Object Model



Manipulating the DOM

Use `document` to reference the DOM.

```
let title = document.getElementById("articleTitle");
let loginBtn = document.getElementsByTagName("button")[0];
let callouts = document.getElementsByClassName("callout"); // *
```

*class refers to a **CSS** class

We can add *event listeners* or read/modify *properties*.

StackBlitz

Manipulating the DOM

Using these DOM elements, we can change the title of the article, add an action for when the button is clicked, and make all of the callouts red.

```
title.innerText = 'My Website!';
loginBtn.addEventListener("click", () => {
  alert("You are advancing to the next part of the site...");
});

for (let callout of callouts) {
  callout.style.color = "red";
}
```

StackBlitz

Badger Recipes

Using HTML, CSS, and JS!

Other Tools

What is this "TypeScript" I hear about?

TypeScript (TS) is a strict syntactical superset of JS developed to enable the development of large-scale applications and to add static typing.

A preprocessor is used to transpile TS to JS.

Safety of Java + Flexibility of JS = TS

We do not cover TS in this course.

What is this "jQuery" I hear about?

A fast, small, and feature-rich JavaScript library.

Contains all of the functions that you wish were in the standard JavaScript library.

- **jQuery**: `$("#login")`
- **DOM**: `document.getElementById('login')`

Keep your jQuery up-to-date!

We do not cover jQuery in this course.

What is this "Bootstrap" I hear about?

A CSS Framework for developing responsive and mobile-first websites.

We *will* cover this in *Web Dev Basics 3* :)

Questions?